Principles of Quantum Mechanics

Part II

Recorded from the lecturer in 2026-2027.

0. Introduction

∗ Recall features from IB QM (Supplementary Notes 1)

- Wave-particle duality:

  Waves behaving like particles, e.g. light quanta, & vice versa (particles behaving like waves).

- Wavefunction $\psi(x)$ for a particle — prob. density $|\psi(x)|^2$.

  (Prob. intrinsic to theory.)

- Observables became operators on wavefns:

$$\hat{x} \to x, \qquad \hat{p} \to -i\hbar \frac{\partial}{\partial x}.$$

  & lack of commutativity, e.g.

$$[\hat{x}, \hat{p}] = i\hbar$$

  leads to Uncertainty Relation.

- Schrödinger eqn with Hamiltonian H specifies dynamics & eigenvalues of H are allowed energy levels of system.

- All of this explains H atom structure & resulting line spectra.

∗ Aims of this course:

- Reformulate QM in a more powerful, flexible, & useful form — the Dirac formalism (go beyond wavefns!).

Previous description appears as particular realisation & also leads to alternatives, e.g.

$$\tilde{\psi}(p) \qquad \text{with} \qquad \hat{x} \to +i\hbar \frac{\partial}{\partial p} \quad \& \quad \hat{p} \to p.$$

This more general approach allows

- Simpler analysis of known problems, e.g. oscillator

but also lots of new things

- Spin of particles
- Symmetries (translations, rotations, for example) & conservation laws.
- Identical particles (bosons & fermions)
- Framework for quantising more general systems, e.g. EM field
- Perturbation theory: practical tool for computations — we will not look at too many detailed applications, but we will try to keep track of what all this is for — physical predictions

1. Dirac Formalism

1.1 Concepts & Definitions

A quantum system is described at each instant by a state $|\psi\rangle$ which belongs to a complex vector space V; any non-zero element of this space V is a possible state.

Thus

$$|\psi\rangle \;\&\; |\phi\rangle \in V \quad \Rightarrow \quad \alpha|\psi\rangle + \beta|\phi\rangle \in V \qquad \text{for any } \alpha, \beta \in \mathbb{C}.$$

Physically this is a superposition principle leading to wave-like behaviour, but these states are not wavefns. There are also dual or conjugate states $\langle\phi|$ which belong to the dual space $V^\dagger$.

By definition, states & duals can be combined or paired to give a complex number

$$\underset{\text{"bra"}}{\langle\phi|}\,, \quad \underset{\text{"ket"}}{|\psi\rangle} \longmapsto \underset{\text{"bra(c)ket"}}{\langle\phi|\psi\rangle}\,, \qquad V^\dagger \times V \to \mathbb{C},$$

with

$$\langle\phi|\big(\alpha_1|\psi_1\rangle + \alpha_2|\psi_2\rangle\big) = \alpha_1\langle\phi|\psi_1\rangle + \alpha_2\langle\phi|\psi_2\rangle,$$
$$\big(\beta_1\langle\phi_1| + \beta_2\langle\phi_2|\big)|\psi\rangle = \beta_1\langle\phi_1|\psi\rangle + \beta_2\langle\phi_2|\psi\rangle.$$

The spaces V & $V^\dagger$ come with an inner product, which can be described as a one-to-one correspondence between the spaces

$$V \longleftrightarrow V^\dagger,$$

with

$$|\psi\rangle \longleftrightarrow \langle\psi| = \big(|\psi\rangle\big)^\dagger \qquad \text{(use same symbol for corresponding states)}.$$

The inner product is

$$|\phi\rangle,\ |\psi\rangle \longmapsto \langle\phi|\psi\rangle = \big(|\phi\rangle\big)^\dagger|\psi\rangle, \qquad V \times V \to \mathbb{C},$$

& it is assumed to obey

$$\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^* \qquad \text{hermitian},$$
$$\big\|\,|\psi\rangle\,\big\|^2 = \langle\psi|\psi\rangle \qquad \text{real},$$
$$\left.\begin{array}{l}\langle\psi|\psi\rangle \geq 0,\\ \langle\psi|\psi\rangle = 0 \quad \text{iff} \quad |\psi\rangle = 0\end{array}\right\} \qquad \text{positive definite},$$

where now $\big\|\,|\psi\rangle\,\big\|$ is called the norm.

Note: Knowing $\langle\phi|\psi\rangle$ for all $\langle\phi|$ determines $|\psi\rangle$ (& vice versa).

[Check:

$$\langle\phi|\psi_1\rangle = \langle\phi|\psi_2\rangle \quad \Rightarrow \quad \langle\phi|\chi\rangle = 0, \qquad \text{where } |\chi\rangle = |\psi_1\rangle - |\psi_2\rangle.$$

If this holds $\forall\,\langle\phi|$, set $\langle\phi| = \langle\chi|$ & get

$$\langle\chi|\chi\rangle = \big\|\,|\chi\rangle\,\big\|^2 = 0 \quad \Rightarrow \quad |\chi\rangle = 0, \qquad \text{as claimed.}$$

]

The physical content of any state is unchanged by

$$|\psi\rangle \to \alpha|\psi\rangle \qquad \text{for any } \alpha \in \mathbb{C}, \quad \alpha \neq 0.$$

We will usually normalise to make $\big\|\,|\psi\rangle\,\big\| = 1$, but can still change

$$|\psi\rangle \to e^{i\theta}|\psi\rangle.$$

Relative phases can be important in expressions like $\alpha|\psi\rangle + \beta|\phi\rangle$.

The space V is complete: we assume that appropriate sequences & series converge. A complete inner product space is called a Hilbert space.

An operator Q is a linear map on states

$$|\psi\rangle \longmapsto Q|\psi\rangle, \qquad V \to V,$$

with

$$Q\big(\alpha|\psi\rangle + \beta|\phi\rangle\big) = \alpha Q|\psi\rangle + \beta Q|\phi\rangle.$$

The same operator can be thought of as acting to the left on dual states,

$$\langle\phi| \longmapsto \langle\phi|Q, \qquad V^\dagger \to V^\dagger,$$

where the right-hand side is defined by

$$\big(\langle\phi|Q\big)|\psi\rangle = \langle\phi|\big(Q|\psi\rangle\big) = \langle\phi|Q|\psi\rangle.$$

End of Lecture 1